Experimental Plan for Spectral Stability of Principal Subspaces

Research design note

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28 October 2026

Abstract

We study finite-sample principal-subspace recovery, the accuracy of first-order spectral risk predictions, and the relevance of spectral diagnostics to equity covariance instability. The design separates known-population estimation error from observable disagreement between estimated subspaces. Hypotheses, controls, outcomes, and interpretation limits are specified before experimentation.

1 Definitions and primary outcomes

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}_d(0, \Sigma)$, with $\lambda_1 \geq \cdots \geq \lambda_d > 0$ and $\delta_k = \lambda_k - \lambda_{k+1} > 0$. Write $P_k = U_k U_k^\top$ and $\hat{P}_k = \hat{U}_k \hat{U}_k^\top$ for the population and estimated top-k projectors. The centered sample covariance uses denominator $m = n - 1$. Define

$$L_k = \frac{\|\hat{P}_k - P_k\|_F^2}{2k} = \frac{1}{k} \sum_{i=1}^{k} \sin^2 \theta_i, \qquad R(\Sigma, n, k) = \mathbb{E} L_k. \tag{1}$$

The Gaussian first-order benchmark is

$$B(\Sigma, n, k) = \frac{1}{k(n-1)} \sum_{i \leq k} \sum_{j > k} \frac{\lambda_i \lambda_j}{(\lambda_i - \lambda_j)^2}. \tag{2}$$

It is a leading-order approximation for fixed d, k, Σ and positive boundary gap as $n \to \infty$, not an exact risk or a universal upper bound. Record both normalized loss and $\|\hat{P}_k - P_k\|_2 = \sin \theta_{\max}$.

2 RQ1: Beyond the boundary eigengap, how strongly does global spectral geometry govern finite-sample subspace risk?

Question. How do n, d, k and global population spectral geometry interact beyond δ_k to affect $R(\Sigma, n, k)$?

Hypotheses. (H1) At finite sample sizes, convergence to the asymptotic $1/(n-1)$ scaling law is governed jointly by the perturbation ratio $\|\hat{\Sigma} - \Sigma\|_2 / \delta_k$ and background spectral mass, rather than sample size alone. (H2) At fixed n, adding dimensions can worsen recovery, but the effect depends on the added eigenvalues. (H3) Equal boundary gaps need not imply equal risk: the full spectrum matters.

Experiment E1: recovery grid. Generate centered Gaussian samples from prespecified spiked and decaying covariance families. Vary $n \in \{100, 250, 500, 1000, 2000\}$, $d \in \{20, 50, 100\}$, and admissible $k \in \{1, 2, 5, 10\}$. Estimate mean L_k, Monte Carlo standard errors, and perturbation ratio $\|\hat{\Sigma} - \Sigma\|_2 / \delta_k$. Use a small pilot before freezing a manageable final grid.

Experiment E2: matched-gap spectra. Construct pairs of spectra with the same absolute or relative boundary gap but different remaining eigenvalues. Control overall scale and, in a separate comparison, effective rank. Compare held-out prediction performance of gap-only, gap-plus-scale, and full-spectrum predictors.

Analysis and decision rule. Report loss-versus-n curves, uncertainty intervals, rank sensitivity, and paired matched-spectrum contrasts. RQ1 is answered only if comparisons identify what was held constant—changing dimension, signal strength, and spectrum simultaneously is not an interpretable isolated effect.

3 RQ2: Where is the breakdown boundary of first-order risk theory, and how does it decouple vector turnover from subspace stability?

Question. Across what regime of noise and dimension does $B(\Sigma, n, k)$ accurately predict R, where does the linear approximation break down, and when does individual vector instability misrepresent subspace stability?

Hypotheses. (H4) B is accurate when sampling perturbations are small relative to spectral separation. (H5) Prediction deteriorates near small boundary gaps or in challenging high-dimensional regimes; the sign of error is not assumed. (H6) Small gaps within the leading cluster destabilize individual eigenvectors without necessarily destabilizing the entire top-k subspace.

Experiment E3: breakdown map. Cross selected n, d, k, spectral families, and relative gaps. For each cell, estimate $\hat{R} = M^{-1} \sum_{r=1}^{M} L_k^{(r)}$ using independent replications. Report $B - \hat{R}$, $|B - \hat{R}|$, and $\hat{R}/B$ when meaningful. Compare discrepancies with Monte Carlo uncertainty; plot heatmaps at fixed d, k and spectrum family. Never clip B to the feasible loss range.

Experiment E4: clustered directions. Use $\Lambda_\varepsilon = (5 + \varepsilon, 5, 5 - \varepsilon, 1, \ldots, 1)$ with $k = 3$. Vary ε while maintaining a separated boundary. Compare ordered eigenvector alignment (only when identifiable) with rank-three projector loss. At $\varepsilon = 0$, individual eigenvector recovery is undefined, not zero.

Mechanism control. Add deterministic perturbations with equal operator norm but different coupling geometry: commuting, single cross-boundary coupling, and multiple couplings. This separates the effect of perturbation direction from its magnitude. A valid Davis–Kahan-type bound may be loose; bound validity and predictive accuracy are distinct questions.

4 RQ3: Can plug-in spectral risk theory predict out-of-sample equity subspace rotation, and when does rotation exceed stationary noise?

Question. Do observable full-spectrum diagnostics—specifically the plug-in estimator $B(\hat{\Sigma}_t, n, k)$—outperform gap-only and historical baselines in predicting rolling equity subspace instability, and when does observed instability exceed a calibrated stationary reference?

Hypotheses. (H7) Smaller normalized sample gaps may be associated with greater subsequent subspace displacement. (H8) Full-spectrum plug-in diagnostics $B(\hat{\Sigma}_t, n, k)$ may outperform simple gap-based and historical baselines. (H9) Conclusions may change under different windows, normalization, asset universes, and stationary-reference assumptions.

Experiment E5: CRSP application. Use documented US common-stock total returns, stable security identifiers, and as-of-date universe formation. Freeze an identical asset coordinate set within each comparison; audit missingness, delistings, and attrition. Estimate covariance projectors in adjacent non-overlapping windows, with overlapping rolling estimates reserved for

descriptive monitoring. Measure

$$S_{t,h}(k) = \frac{\|\hat{P}_{k,t+h} - \hat{P}_{k,t}\|_F^2}{2k}. \tag{3}$$

Compare gap-only and full-spectrum diagnostics, including the plug-in $B(\hat{\Sigma}_t, n, k)$, against a simple historical baseline using chronological held-out periods. Evaluate a whole-vector stationary block-bootstrap reference, only after checking its calibration on synthetic stationary and changed-covariance data.

Interpretation limit. $S_{t,h}$ is observable disagreement, not true population recovery error. A bootstrap-reference exceedance does not prove a regime change: volatility shifts, dependence, selection, and estimation noise can all contribute. If calibration fails, report descriptive comparisons without inferential claims.

5 Validation, outputs, and execution

For synthetic cells, report means, Monte Carlo standard errors, and intervals; use independent random streams and prespecified seeds. Test numerical projector identities and exact deterministic rotations. Targeted robustness includes heavy-tailed returns, altered k, alternative window lengths, correlation PCA, and an estimator sensitivity check. Do not interpret repeated overlapping windows as independent observations.

The minimum paper figures are: (i) risk versus sample size; (ii) matched-gap comparisons; (iii) predicted versus simulated risk; (iv) benchmark-error map; (v) vector versus subspace instability; and (vi) empirical disagreement with calibrated references. Every figure must answer a stated hypothesis.

Execution order. Freeze hypotheses and controls; implement and test projector metrics; run E1–E2; run E3–E4; validate uncertainty and empirical reference models; conduct E5; synthesize results with negative findings retained. The research succeeds if it identifies where the approximation is informative, where it fails, and what can legitimately be inferred from equity data—not if every hypothesis is confirmed.

References

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[3] D. Politis and J. Romano, "The stationary bootstrap," *JASA* 89 (1994), 1303–1313.